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Enhancing plant stress diagnosis using thermal imagery and global context vision transformer with explainable AI approach

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Abstract
The combination of artificial intelligence (AI) and computer vision is transforming agriculture by offering information that helps to optimize crop and resource management and, thus, boost productivity. The early warning on the occurrence of plant stress due to various environmental factors is through AI, and it helps reduce losses in yield. Thermal imaging is used to record temperature difference in vegetation and identify different physiological stresses, which can be used to prevent serious damages by agronomists before they take place. The study examines the deep learning algorithms to detect plant stress by detecting thermal images of paddy leaf blades. Although convolutional neural networks (CNNs) have long been the most popular in the image classification task, Vision Transformers (ViTs) are proving to be an effective alternative. This study assesses various CNN models, including VGG-16, VGG-19, ResNet-50, and Xception, as well as the Global Context Vision Transformer (GCViT). The findings reveal that ResNet and GCViT models outperform other models, demonstrating higher accuracy with variation of around 3% accuracy across various dataset folds and classes. This work introduces a comparative analysis of several CNN structures (VGG-16, VGG-19, ResNet-50, Xception) and the GCViT. ResNet and GCViT will be shown as the most accurate (0.93–0.97) and stable in various dataset folds and classes. Also, explainable AI is applied to the model to identify the part of the image that decides the stress class.

Article Highlights

- The study helps identify the type of stress in plants by employing computer visionand deep learning methodologies.
- This research investigates the efficacy of different convolutional neural network(CNN) models in conjunction with the Global Context Vision Transformer (GViT)architecture.
- Explainable AI is used to increase trust in decisions made by deep learning models.
- The study will help to develop a non-invasive system for early detection of plantstress to improve crop yields.

Keywords Convolutional neural networks, Deep learning, Explainable AI, Global context vision transformer, Plant stress, Thermal imaging

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1 Introduction

Recent research highlights the potential of deep learning in transformation of various sectors, including agriculture [1]. As the global population grows rapidly, agricultural production encounters significant challenges to fulfil demand [2, 3]. Effectively addressing the issue of reducing yield losses due to various forms of plant stress is essential. Application of deep learning methods in detecting stress in plants is a challenging research problem [4]. Agricultural productivity can be improved through the early detection of plant stress, the application of various imaging modalities with deep learning networks [5, 6].

CNNs prove better than alternative deep learning approaches at image classification through their exceptional performance levels. The CNN framework contains multiple architectural designs that users can investigate. The research evaluates the performance output of VGG-16, VGG-19, ResNet-50 and Xception network architectures. The Visual Geometry Group at Oxford University created VGG-16 as a CNN with 16 layers known as VGG-16. Its counterpart, VGG-19, features a total of 19 layers. Microsoft Research created ResNet-50 as a residual network that includes 50 layers for processing. The network has established a reputation by providing efficient image classification while maintaining a deep architecture. Xception is a 71-layer CNN architecture incorporating depthwise and pointwise convolution in its operation. The model originated from François Chollet, who worked at Google.

The research evaluates four recognised CNN models, which include VGG-16 and VGG-19 as well as ResNet-50 and Xception. All of these models demonstrate particular advantages that set them apart. The VGG-16 and VGG-19 architectures feature basic design simplicity, enabling effective, detailed feature extraction from images. ResNet-50 builds its deep architecture around residual connections, preventing typical training problems affecting deep networks. The convolution method employed by Xception represents its main enhancement over previous models to produce faster processing time during image analysis. Testing involves studying both CNNs and Vision Transformers with GC mechanisms included within their structures. The attention mechanism that ViTs have improves understanding of the relationship between pixels, whereas CNNs mainly detect local spatial variations. This paper analyses various models to establish which one works best in the analysis of paddy leaves thermal images.

Stress indicators are especially difficult to extract out of thermal images as they are subtle, and widely spread out in the image. This paper introduces a modified version of GCViT that uses thermal imaging to identify plant stress. It addresses the limitations of traditional CNN techniques, which often overlook long-range thermal interactions. GCViT combines global self-attention with local details, making it sensitive to small temperature changes caused by stressors like drought or nutrient deficiency. This improves its accuracy in detecting identifying type of stress. Unlike earlier adaptations of ViT that focused on RGB images, this paper is the first to apply GCViT to thermal image data. It shows strong performance across different crop types and environments, supported by data augmentation. The model's efficient computing capabilities allow it to be used on edge devices, enabling real-time field monitoring to support sustainable agriculture. This research connects computer vision and farming, providing a scalable and non-invasive tool for precision agriculture.

The organization of the paper is as follows: Sect. 2 provides the literature survey discussing the models like CNN, ViT, and GCViT in the framework of plant stress and disease detection by analysing images. Section 3 provides the comprehensive description of the ViT and GCViT structures utilized to classify images. Finally, Sect. 4 provides a summary of the empirical findings that were made on all the models and provides final observations regarding these findings.

2 Related work

Plant stress can be attributed to various causes, which are primarily classified into two categories: biotic and abiotic stress. Biotic stress originates from living organisms, including insects, weeds, fungi, viruses, and bacteria [7]. In contrast, abiotic stress arises from nonliving factors that disrupt the normal growth and development of plants [8]. Several technological developments in recent years are transforming the technology of plant stress detection. The stress detection methods carried out by imaging give a number of advantages, including the possibility to determine the stages of stress, the non-destructive character of monitoring via drones or satellites, and the possibility to cover extensive areas [9]. Also, these imaging technologies allow real-time stress detection and can even fully automate the stress monitoring process [10].

The adverse effects of stress in plants are a decreased cell division and elongation leading to decreased growth. It also introduces significant delays in flowering and fruit stages in plants. To be precise, the stress caused by water may result in dehydration or wilting of plants [11]. Besides, chlorophyll content also decreases with stress and this decreases the ability of the plant to absorb light and fix carbon. Plants lose more energy as the level of respiration rises as a result of stress. Moreover, it can result in the flowering of flowers drop before they mature [12].

The effects of stress on plants are different based on the type of stress which may be biotic or abiotic stress and subtype stresses [13]. The abiotic stressors (drought) affect transpiration and photosynthesis negatively and salinity stress affects the balance of cellular ions and nutrients in plants [14]. Also, the effect of temperature stress can be protein denaturation and rigidity of cell membranes.

On the other hand, biotic stress is caused by living organisms that may cause damage to plants physically and a nutritional deficit [15]. As an illustration, parasitism weeds may greatly lower the water and nutrient supply of host plants. Leelavathy and Reddy [16] uses the VGG-16 model in detecting plant diseases with an amazing accuracy of 93%. In the meantime, Azimi et al [17] applies a CNN with LSTM (CNN-LSTM) to identify water-stress in a chickpea dataset, achieving an accuracy of 98.52% in the JG-62 variety and 97.78% in the Pusa-372 variety. Syihad et al. [18] used CNN models to detect diseases in banana plants by examining the images of banana leaves. The ResNet-50 model demonstrated a high level of accuracy of 94%, precision, and recall of 88% and 91% respectively. The VGG-19 model also had a 91% accuracy of detecting diseases of banana plants. In another paper by Kamarudin et al. [19], a lightweight CNN model was applied to measure stress of Setaria grass under three stress levels of water. The results showed a greater improvement of more than 9% in accuracy, F1 score, recall and precision to a other CNN model.

In a similar study, Khalifani et al. [20] applied a CNN to sunflower grains to identify salinity stress. They conducted a comparative analysis between an ANN and CNN

models, finding that the CNN model outperformed the ANN model in detecting both typical and stressed plants. In another research work, Deng et al. [21] utilised CNN models to evaluate salt stress levels in soybean seedlings. They compared several CNN architectures, including AlexNet, GoogLeNet, ResNet-50, ShuffleNet, SqueezeNet, and MobileNetv2. By analyzing a dataset of chlorophyll fluorescence images, they discovered that ResNet50 achieved an impressive accuracy of 96.23% on the test set and 96.06% on the validation set, surpassing the performance of the other models in detecting salt stress levels.

Similar research was done by Khalifani et al. [20], who used a CNN to sunflower grains to detect salinity stress. A comparative study of ANN and CNN models was carried where they discovered that CNN model was superior to ANN model in the detection of typical plants and stressed plants. Deng et al. [21] apply CNN models in a separate study to estimate the level of salt stress in soybean seedlings. They benchmarked a number of CNNs, among them, AlexNet, GoogLeNet, ResNet-50, ShuffleNet, SqueezeNet, and MobileNetv2. Through their analysis, they found that ResNet50 attained a high accuracy of 96.23% and 96.06% on the test set and validation set, respectively, which was better than the performance of the other models in predicting the level of salt stress.

In one of the recent studies, Patra et al. [22] applied a vision transformer algorithm in a combination with a SVM to detect drought stress in potato plants. The study authors used a data set of aerial photographs and compared the results of ViT + SVM with pre-trained weights, optimum weights, and a ViT model trained on transfer learning. Their results revealed that the maximum accuracy of 92.95% was obtained when using ViT model, transfer learning. Another research by Thokala and Doraikannan [23] examined a Deep CNN that is combined with a multi-scale ViT (DCNN-MCViT). This model was better than other models when it comes to comparison to other CNN architectures, and it has an amazing rate of accuracy of 99.51% at recognising different forms of stress. Also, Lehouel et al. [24] made a comparative study of independent ViT and CNN-ViT on satellite images to detect water stress on crops. ViT in isolation got the accuracy of 88% and CNN-ViT got a bit higher with the accuracy of 90%.

The literature review points out that it is critical to examine the identification of the plant stress by the use of ViT models. The most recent studies in the field were selected and the key findings are summarized in the Table 1 that also gives the choice of the methodology, datasets and performance indicators.

3 Methodology

This study has analyzed a plant stress detection model using ViT. There are several approaches to measuring the stress of plants, but in this study, the use of the ViT-based approach is particularly highlighted.

3.1 Vision transformer

Computer vision is increasingly employed across various applications to automate systems and minimize manual errors stemming from human limitations in image storage and processing, particularly within agricultural contexts [30]. A highly accurate computer vision system can significantly improve the effectiveness of plant stress detection through image analysis. The ViT is a specialized architecture created for computer vision applications. In recent years,CNNs have primarily led the field of computer vision,

Table 1 Summary of recent studies employing deep learning models in plant stress and disease detection

Study	Techniques used	Datasets employed	Results achieved	Identified limitations
Ghazal et al. [25]	Fine-tuning ViT-B 16 to Vision Transformer Custom-built Vision Transformer (ViT), fine-tuning CNN models (ResNet50, EfficientNetB0, InceptionV3, DenseNet121) with transfer learning	Mendeley maize RGB images (1200 images, 3 classes: N0, N75, NFull).	ViT Fine-tuned EfficientNetB0, 97% test accuracy; custom ViT 76%; fine-tuned ViT 64%	ViTs also meet prematurely with greater losses, need more samples of data, and need more training time and parameters, which are less applicable to real-time agricultural applications.
Barman et al. [26]	Vision Transformer (ViT), Inception V3	PlantVillage tomato leaf subset 10,010 images (10 disease groups and control, 10,010 images)	ViT: 90.99% testing accuracy 97.37% training, 95.76% validation Inception V3: 94.80% training, 94.02% validation	Coverage should be restricted to classification; object detection architecture of weeds and plants will be needed in the future.
Thakur et al. [27]	PlantXViT: ViT-enabled CNN combining VGG16, Inception block, and transformer encoders with multi-head self-attention	Five publicly available datasets, which range in size from small, balanced sets with 400–500 images to huge, imbalanced sets with over 40,000 images.	Accuracies: 93.55% (Apple), 89.24% (Embrapa), 92.59% (Maize), 98.86% (PlantVillage), 98.33% (Rice); outperformed state-of-the-art CNNs	Inception block with higher computational requirements (GFLOPs), which slows applicability to the IoT; intend to lower FLOPs.
Reno et al. [28]	Multilayer Perceptron (MLP) and CNNs for feature extraction and classification from thermal images	Using the FLIR Thermal Camera T560, 419 thermal images of 27 cotton genotypes in progressive drought	MLP (212 well-watered, 207 dry-down) yielded 78% accuracy; better than random forest (75%) data.	Low image quality and blurriness with hand-held cameras; genotypic differences and microclimate influence causing misclassifications.
Ashurov et al. [29]	Depthwise CNN; depthwise separable convolutions; L2 regularization (modifications to MobileNetV2)	publicly available dataset, which contains RGB images	Accuracy: 98%; F1-score: 98.2%; precision: 97%; recall: 99%; inference time: 12 ms per image; model size: 8.5 MB;	Data imbalance that needs more effective augmentation; model interpretability issues.

especially in image classification tasks. Known for their superior accuracy compared to other algorithms of that era, CNNs have established themselves as the state-of-the-art method in this domain [31].

The transformer model is commonly used for NLP tasks involving text. The ViT model, on the other hand, processes images in a manner similar to how words are handled in NLP with transformers [32]. ViT begins by dividing an image into small patches, typically sized at 16×16 pixels. These patches are then flattened into vectors to maintain the positional information of each patch, and positional encoding is applied [33, 34]. Positional encoding is crucial as it enables the model to identify the exact location of each patch within the original image, which is important for recognising spatial relationships.

The multilayer transformer plays a vital role in the ViT architecture [35]. It consists of two main components: the attention layer and the feed-forward layers. The self-attention layers help the model understand the relationships between different patches, while the feed-forward layer applies a non-linear transformation to identify complex patterns in the images [36].

3.2 Global context vision transformers

ViT model has limitations, like it requires a vast dataset size to perform well. ViT performance decreases on local detail capturing. The above-mentioned limitations are removed in the GCViT [37]. The architecture of the GCViT model is shown in the Fig. 1.

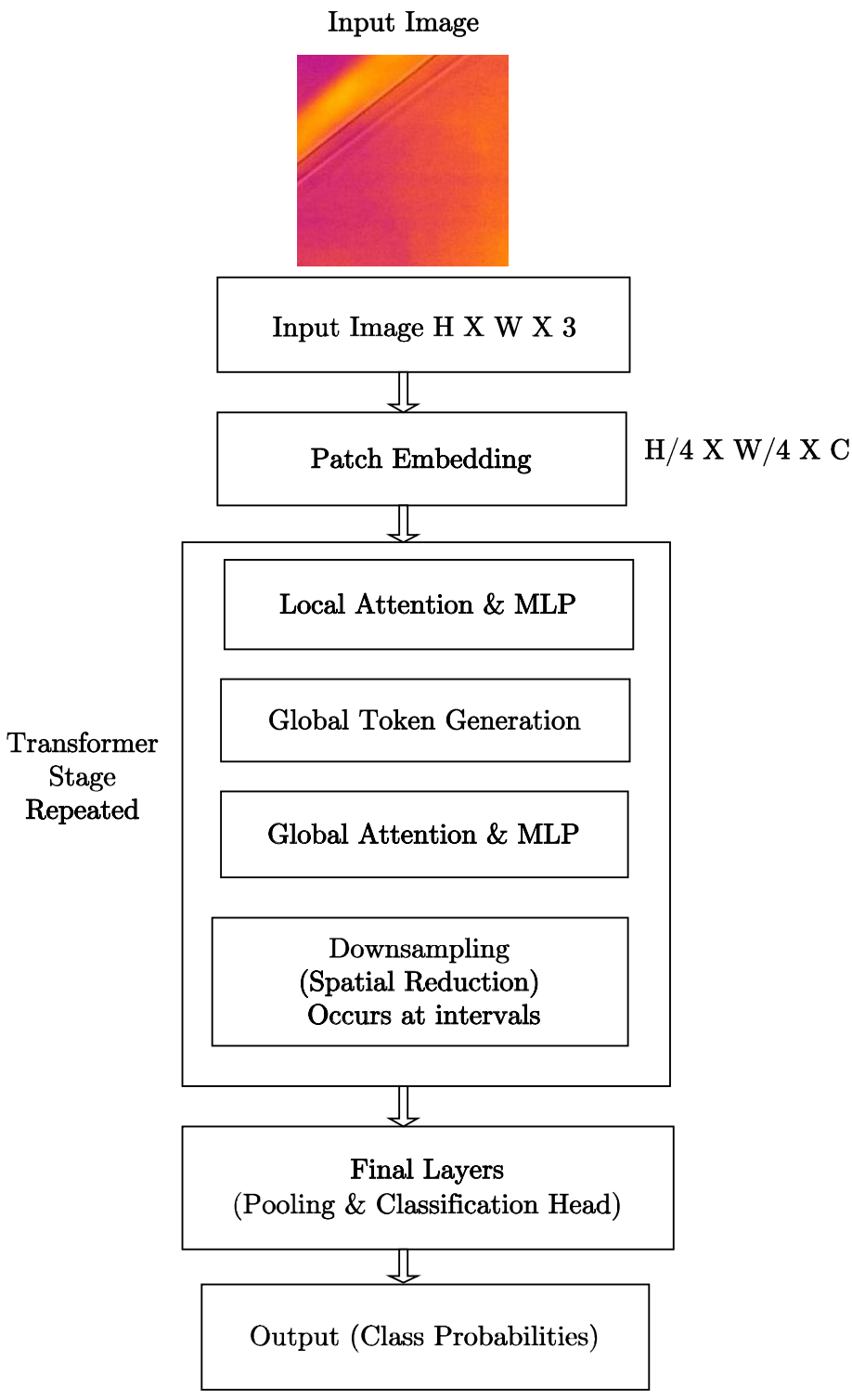


Fig. 1 Architecture of GCViT model

Thermal images processed as a three-channel pseudo-color image of width W and Height H are given as input to the GCViT model. In the GCViT Stem model, the image is divided into small patches. These patches are then converted to the fixed size of feature space. Each Patch is linearly transformed in to vector of size C. Due to this step, spatial dimension was reduced to $H/4 \times W/4$, and feature depth increased to C.

The input image $X \in \mathbb{R}^{H \times W \times 3}$ is divided into patches, and each patch is flattened and linearly projected into a higher-dimensional space:

$$T_0 = \text{Linear}(\text{Patchify}(X)) \quad \text{where} \quad T_0 \in \mathbb{R}^{N_0 \times C_0} \tag{1}$$

Here:

$N_0 = \frac{H}{P} \times \frac{W}{P}$: Total number of patches, where P is the patch size.

C_0: Embedding dimension of each patch token.

GCVit architecture has multiple layer arrangements that refine image features step by step. Each level consists of the following components:

A. Local MSA (Multi-Head Self-Attention): This module examines how local features connect to each other inside specific image parts. It uses limited areas of the image to run calculations faster.

The self-attention within local windows is computed as:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V \tag{2}$$

where:

$Q = T_0 W_Q, K = T_0 W_K, V = T_0 W_V$

$W_Q, W_K, W_V \in \mathbb{R}^{C_0 \times d_k}$: Learnable projection matrices.

d_k: Dimensionality of the attention head.

B. Global MSA (Multi-Head Self-Attention): Global Token Generation module combines global token insights to enhance attention processing. The model learns image context by using global tokens to capture both nearby and distant relationships between image elements.

To capture global context, global tokens are generated and incorporated into the attention mechanism.

1. Global Token Generation:

$$G = \text{Pool}(T), \quad G \in \mathbb{R}^{N_g \times C_g} \tag{3}$$

where:

G: Global tokens summarizing spatial features.

N_g: Number of global tokens.

C_g: Embedding dimension of global tokens.

2. Global Attention:

$$\text{Attention}_{\text{global}}(Q, K, V) = \text{Softmax}\left(\frac{Q \begin{bmatrix} K_T \\ K_G \end{bmatrix}^\top}{\sqrt{d_k}}\right) \begin{bmatrix} V_T \\ V_G \end{bmatrix} \tag{4}$$

where K_T, K_G and V_T, V_G represent keys and values for spatial and global tokens, respectively.

C. MLP (Feed-Forward Network): After self-attention processing the MLP (Multi-Layer Perceptron) improves non-linear feature transformations.

The feed-forward network is applied as:

$$T' = \mathrm{MLP}(T) = \sigma(TW_1 + b_1)W_2 + b_2 \tag{5}$$

Here:

σ: Non-linear activation function (e.g., GELU).

W_1, W_2: Learnable weight matrices.

b_1, b_2: Bias terms.

D. Downsampling: At each stage, the network downsamples feature maps to reduce their spatial resolution while expanding the number of feature channels to preserve information content. The spatial resolution of tokens is reduced using a convolutional layer:

$$T' = \mathrm{Downsample}(T) = \mathrm{Conv2D}(T) \tag{6}$$

Next step is Global Token Generation, in this step, to process global context information, a Token Generation system creates summaries at every network layer. The network uses global tokens as input to attention layers, which keeps global information present across all network levels. In the Feature Extraction step, as the network advances through stages, it extracts detailed feature information by decreasing spatial size and expanding feature complexity. Through this design, GcViT handles large input data at high resolution by keeping both local details and global context. At each hierarchical stage, the features are updated using multi-head self-attention and an MLP:

$$T^{(l+1)} = \mathrm{Downsample}(\mathrm{MSA}(\mathrm{MLP}(T^{(l)}))) \tag{7}$$

In the Final Layers of GCViT, the network uses 2D Average Pooling to analyse the final feature map after completing all its hierarchical levels. The combined features move to the classification head, which creates the final result. The final feature map is globally pooled and passed to the classification head:

$$y = \mathrm{Head}(\mathrm{GlobalPool}(T)) \tag{8}$$

Here:

$\mathrm{GlobalPool}(T)$: Aggregates spatial tokens into a single vector.

Head: Linear classification head.

y: Final prediction output.

3.3 Summary of notations

H, W: Height and width of the input image.

P: Patch size for dividing the image.

N_0, C_0: Number of patches and embedding dimensions of tokens.

Q, K, V: Query, key, and value matrices in self-attention.

d_k: Dimensionality of attention head.

G: Global tokens summarizing spatial features.

N_g, C_g: Number and embedding dimensions of global tokens.
$T^{(l)}$: Tokens at hierarchical stage l.
σ: Non-linear activation function.
y: Final output of the classification head.

The GCViT takes input images and performs a task, like classification, by combining local and global attention feedback mechanisms to efficiently extract short and long-range spatial dependencies. GCViT starts with the input image of size $\mathrm{H} \times \mathrm{W} \times 3$, which is partitioned into non-overlapping patches and encoded linearly into tokens of size $\mathrm{H}/4 \times \mathrm{W}/4 \times \mathrm{C}$, initial spatial reduction is introduced. Transformer stages are then repeated in the model. It does this by first self-attending on the local window in every stage, and then a MLP to approximate small-range interactions between neighboring tokens. Then, GCViT produces global tokens which represent a summary of information over an image, making it possible to do long-range modeling. It then performs a global self-attention step, in which global queries engage with local keys and values, and then another MLP, to add cross-region context without expensive operations such as window shifting or attention masks. The model performs downsampling through spatial reduction methods, i.e. patch merging, at predefined intervals between stages, to reduce the number of tokens and computational load whilst maintaining features. In place of inductive bias on the final block, modified fused inverted residual blocks introduce inductive bias at each of the aforementioned stages, improving representation over conventional Vision Transformers. Lastly GCViT bunches the processed tokens and feeds them through a classification head to generate class probabilities. By incorporating global self-attention methods that capture long-range dependencies that CNNs limit through local receptive fields, GCViT outperforms CNN models. By switching between local and global attention, its hierarchical architecture improves feature representation at all resolutions. Compared to pure CNNs, GCViT improves efficiency and parameter utilization by incorporating convolutional inductive biases through downsampling blocks.

3.4 Explainable AI

Explainable Artificial Intelligence (XAI) shows promise in improving precision agriculture and plant phenotyping by using thermal imaging to detect stress in plants. Plants emit infrared energy which this technology measures to deliver non-harmful assessments of their biological signaling processes including transpiration as well as stomatal conductance and water condition. Thermal image analysis detects important plant stress indicators that result from various environmental stressors including drought conditions and nutrient deficiencies and pathogen infections. Thermal imaging data presents a main obstacle because most AI models operate with unidentified processes within their systems. Black box operation hampers scientists' ability to trust the predictions made by these systems. XAI delivers explanations for AI model outcomes by providing transparent reliable and understandable explanations which address the problems of mistrust.

XAI improves AI system transparency which provides human users understanding of the decision processes. The utilization of XAI techniques analyzes which thermal image features lead to plant stress detection outcomes. This understanding allows researchers to compare model predictions with known physiological systems, which helps verify and improve the model's reliability.XAI provides essential functionality to deep learning systems because it enables users to believe in model-based decisions. Through XAI

researchers obtain clarity about what makes models arrive at their chosen outcomes. Image-based applications benefit from XAI by identifying the important image elements that ensure effective decision making. Deep learning models routinely train for plant stress detection through the use of plant leaf images. Through XAI proposed method receive important data about the leaf features which drive model choices. The attention mechanism of ViT plants stress detection methods allocates importance ratings through a weighing system to various image regions. Different sections of images receive weights based on their significance. Attention-based visualisation uses weights from the attention layer to create heatmaps that indicate important image areas responsible for decision outputs.

Heatmaps are usually created using XAI methods such as Grad-CAM, visualizing areas in plant imagery that most affected the deep learning model's predictions. By computing the gradients of class scores and superimposing the activation patterns, these heatmaps point to specific plant stress markers, such as lesions and wilting, allowing precise localisation of affected sites. Furthermore, heatmaps can help diagnose model errors by showing whether attention is leaking to non-relevant objects and help identify appropriate dataset/architecture changes. In academic and research settings, they can help clarify the functioning of stress physiology and connect model focus with biological phenomena such as chlorophyll degradation under drought conditions. Additionally, because of their visual nature, complex deep learning outputs are more approachable even to non-specialists which facilitates decisions for precision farming related to irrigation and pest control.

Each image region gets weights that are proportional to their relative importance. Attention visualization builds heatmaps based on the attention layer weights, thus identifying the parts of the image that play a significant role in the decision-making process of the system. The 'GCViT block' was set up to end at a layer 'layers.multiply' where representation of features altered by the channel-attention mechanism were recorded. Gradients of the target class score, in terms of this particular layer, were then calculated with the help of the Grad-CAM method. The resultant map is class-discriminative, which is, in effect, visualization of the exact spatial regions that the model focused on after the use of the attention mechanism.

3.5 Dataset

This study uses paddy leaves open-source dataset which is available in the Kaggle platform. The FLIR E8 thermal imaging camera was used in recording these images; the camera had a resolution of 320 × 240 pixels. The dataset consists of five categories, one of which is Bacterial Leaf Blight, another one is Blast, Leaf Spot, Hispa, and Healthy Leaves [38].

This dataset was augmented in terms of image augmentation, involving horizontal and vertical flipping, 45-degree and 90-degree rotations. After these augmentation processes the images were scaled to 224 × 224 pixels.

As a result of the augmentations, the total number of images increased to 2,200. The dataset was then divided into three subsets for training(1408 images), testing(440 images), and validation(352 images). The organisation of the dataset is structured into three directories, labelled "Train," "Test," and "Validation." Sample images from dataset are shown in Fig. 2.

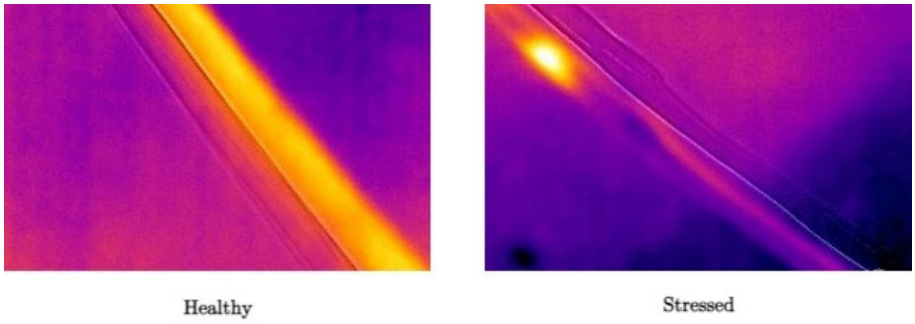


Fig. 2 Sample images from dataset

This research work utilized K-fold validation. To facilitate this, a modified version of the dataset was created that omitted the original training–testing-validation separation. This revised dataset is organized into five distinct subfolders, each representing one of the following class types: Blast, Bacterial Leaf Blight, Healthy, Hispa, and Leaf Spot.

In order to maintain consistency of the experiment, the amount of data that was distributed to each split was held constant throughout the experiment; the only difference was the composition of the datasets themselves as the samples were swapped among the folds.

The distribution of images within each class is as follows: the Blast class contains 469 images, the Bacterial Leaf Blight class has 440 images, the Healthy class includes 465 images, the Hispa class comprises 426 images, and the Leaf Spot class contains 400 images.

4 Results

For this study Google Colab used, a Cloud-based tool for running our code. In Colab, NVIDIA Tesla T4 GPU is used, which provides up to 16 GB of memory of GGDR6 type [39].

A fixed random seed of 42 was used to standardize the model configuration and training process in order to ensure reproducibility. The network takes input measurements of images of size 224 x 224 pixels which are scaled to the range [0, 1] and are carried out in batches of 32. The architecture is based on a hierarchical feature extraction model, where the initial convolution and following GCViT blocks are based on 64, 64 and 128 filter sizes respectively. This step of extracting the features is then followed by a dense layer with 512 units with a dropout rate of 0.5, which allows the reduction of overfitting and the ultimate classification.

As shown in Fig. 3 graph of accuracy over Epochs and Loss over epochs. The “Accuracy over Epochs" graph depicts the progression of training and validation accuracy as the number of epochs increases. Generally, this accuracy tends to improve with additional epochs. Conversely, the “Loss over Epochs" graph reflects the changes in training and validation loss over the same number of epochs; ideally, this loss should decrease as the epochs progress. In this research, every model and each dataset fold was trained for a total of 100 epochs.

The confusion matrix is another fundamental tool of evaluation of performance that is used to estimate accuracy, precision, recall, and F1 score [40]. It is used as a basic way of measuring how effective classification models are, particularly in deep learning systems.

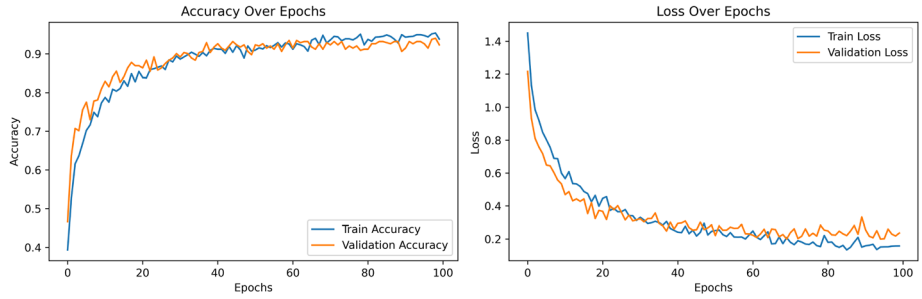
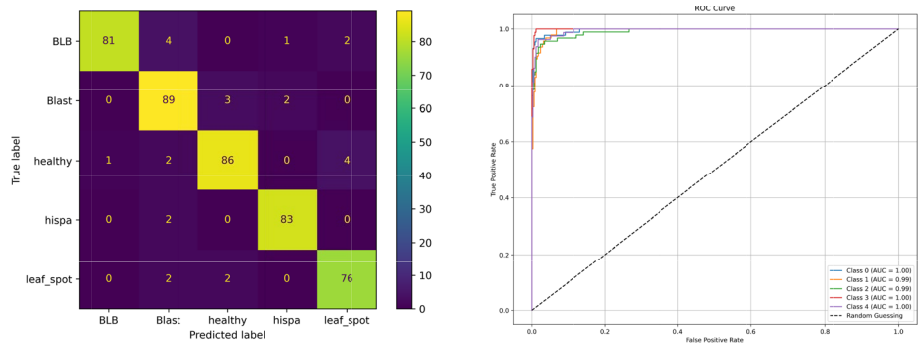


Fig. 3 Sample of accuracy and loss over epoch curve



(a) Sample of the confusion matrix. (b) Sample of ROC Curve.

Fig. 4 Sample of confusion matrix and ROC curve

In contrast to accuracy scores, the confusion matrix is more informative as it indicates the tendencies of misclassification and offers specifics on how each of the classes has performed [41]. The sample of one of the confusion matrices that will be used in this study is presented in Fig. 4a.

The classification thresholds of ROC give an output in the form of a graphical representation indicating various combinations of True Positive Rate (TPR) and the False Positive rate (FPR) [42]. When the values of the AUC are close to 1.0 the performance of better classification is attained. The method offers an applicable analysis of skewed data when precision does not give significant information. Deep learning models that involve the use of CNN, ViTs and Transformers in multi-class classification can obtain the ROC curves on a per-class basis, by computing their one-vs-rest (OvR) curves. An example of the ROC curve is given in Fig. 4b.

The given dataset incorporates the performance indicators of different machine learning models: VGG-16 and VGG-19, ResNet-50, Xception, and GCViT. These models are compared to various folds and classes with regard to various criteria: precision, recall, F1-score, support, accuracy, macro average, and weighted average. A study of the efficiency of each model in data classification into five different categories, including; 'BLB', 'Blast', 'healthy', 'hispa', and 'leaf spot'.

Using VGG-16, the model has shown the same level of performance on all folds and obtained accuracies ranging between 0.90 to 0.94. The macro average F1-scores fluctuate slightly, ranging from 0.91 to 0.94, while the weighted average F1-scores show a similar range. When examining individual classes, 'hispa' consistently receives high precision

and recall scores, whereas 'leaf spot' exhibits relatively weaker performance, particularly in fold 3 and fold 4.

This low sensitivity to the leaf spot category of VGG16 model is probably due to the large quantity of Max Pooling used in the architecture, which aggressively reduces the spatial resolution. This procedure has a tendency of blurring the fine-grained, high-frequency resolution of small localized lesions. Moreover, with the increasing size of the effective receptive field in deeper layers to acquire global context, the local signal of small spots can be masked by the large background features of the health leaf tissue.

This indicates that VGG-16 handles certain classes more effectively than others, which is an important consideration for practical applications requiring balanced classification across all categories. Table 2 shows results for VGG-16 from fold 1 to fold 5.

In contrast, VGG-19 shows lower overall performance compared to VGG-16. Its accuracies are 0.87, 0.90 while the macro and weighted average F1-scores are also in the same range. It is important to note that the values of precision and recall vary across the various folds suggesting a compromised performance stability. Classes like the healthy and hispa classes are moderate to high performing and there is a lot of disparity in others like the 'BLB' and 'Blast'. Such discrepancies point to the possible limitations of VGG-19 to process different datasets. Results of VGG-19 fold 1 to fold 5 are indicated in Table 3.

ResNet-50 has been proven to be of high-performance with accuracies of well over 0.92. The macro and weighted average F1 scores tend to be higher than 0.93 and this indicates the reliability of the scores. ResNet-50 performs best in the classification of the categories: 'BLB', 'Blast', and 'hispa', in which both the precision and the recall scores are high. ResNet-50 still delivers suitable performance even in more difficult scenarios such as 'leaf spot', which makes it a reliable option when it comes to classification tasks with high accuracy and stability. Table 4 presents the ResNet-50 outcomes in fold 1 to fold 5.

Xception is also effective, but it demonstrates a minor lack of consistency in relation to ResNet-50. It has accuracies between 0.91 and 0.95 and macro and weighted average F1-scores are typically between 0.91 and 0.95. Xception is performing well in the category of the 'BLB' and 'hispa', but experiences some variations in the'Blast' and the 'leaf spot' category. Irrespective of such minor irregularities, Xception is a competitive model particularly in cases whereby adequate computing resources are at hand. Table 5 demonstrates that the results of Xception in fold 1 to fold 5 have been taken.

Finally, GCViT can be considered one of the best-performing models, having outstanding metrics in all folds. It has a range of accuracies between 0.93 and 0.97 and macro and weighted average F1-scores are frequently greater than 0.94. GCViT has a specific high performance in the categories of BLB', 'Blast', and 'hispa' where the precision and recall are high and steady. Although the traditionally problematic classes, such as the ones related to the terms 'healthy' and 'leaf spot', have high scores, it confirms the appropriateness of GCViT to the complex classification tasks. Figure 5 is a summary of the accuracy of all the models from fold 1 to fold 5. Table 6 presents the performance of the GCViT model using evaluation metrics across five folds from fold 1 to fold 5.

In this study, Explainable AI is used to detect the critical regions in images of plant leaves. Since deep learning models are often viewed as black boxes, it may be very difficult to comprehend the mode by which they reach their ultimate judgments. Thus, Explainable AI is needed to create confidence in the findings of such models. In the given paper, attention-based Explainable AI is added to the GCVit Model. Attention

Table 2 Performance Metrics for VGG-16 - Fold 1 to Fold 5

VGG-16 - Fold 1				
Class	**Precision**	**Recall**	**F1-Score**	**Support**
BLB	0.86	0.95	0.90	88
Blast	0.91	0.91	0.91	93
Healthy	0.92	0.92	0.92	93
Hispa	0.99	0.97	0.98	86
Leaf Spot	0.94	0.85	0.89	80
Accuracy	0.92			
Macro Avg	0.93	0.92	0.92	440
Weighted Avg	0.92	0.92	0.92	440
VGG-16 - Fold 2				
BLB	0.89	0.98	0.93	88
Blast	0.97	0.90	0.93	94
Healthy	0.94	0.95	0.94	93
Hispa	0.93	1.00	0.97	85
Leaf Spot	0.99	0.86	0.92	80
Accuracy	0.94			
Macro Avg	0.94	0.94	0.94	440
Weighted Avg	0.94	0.94	0.94	440
VGG-16 - Fold 3				
BLB	0.89	0.92	0.91	88
Blast	0.86	0.97	0.91	94
Healthy	0.83	0.91	0.87	93
Hispa	1.00	0.87	0.93	85
Leaf Spot	0.97	0.80	0.88	80
Accuracy	0.90			
Macro Avg	0.91	0.89	0.90	440
Weighted Avg	0.91	0.90	0.90	440
VGG-16 - Fold 4				
BLB	0.91	0.95	0.93	88
Blast	0.92	0.89	0.91	94
Healthy	0.84	0.94	0.89	93
Hispa	0.98	0.95	0.96	85
Leaf Spot	0.92	0.81	0.86	80
Accuracy	0.91			
Macro Avg	0.91	0.91	0.91	440
Weighted Avg	0.91	0.91	0.91	440
VGG-16 - Fold 5				
BLB	0.96	0.93	0.95	88
Blast	0.90	0.95	0.92	94
Healthy	0.98	0.89	0.93	93
Hispa	0.95	0.96	0.96	85
Leaf Spot	0.91	0.96	0.93	80
Accuracy	0.94			
Macro Avg	0.94	0.94	0.94	440
Weighted Avg	0.94	0.94	0.94	440

Heatmap is a type of insight that gives the information on the part of the image that is given more attention than others. The original images and the Attention Heatmaps are provided in Figs. 6, 7, 8, 9, 10.

The integration of XAI into our plant stress detection pipeline—specifically through the use of heatmap visualizations from the GCViT model—has yielded valuable insights into the relationship between thermal anomalies and different types of stress in crops.

Table 3 Performance Metrics for VGG-19 - Fold 1 to Fold 5

VGG-19 - Fold 1				
Class	**Precision**	**Recall**	**F1-Score**	**Support**
BLB	0.87	0.85	0.86	88
Blast	0.94	0.84	0.89	93
Healthy	0.91	0.92	0.92	93
Hispa	0.91	0.95	0.93	86
Leaf Spot	0.83	0.90	0.86	80
Accuracy	0.89			
Macro Avg	0.89	0.89	0.89	440
Weighted Avg	0.89	0.89	0.89	440
VGG-19 - Fold 2				
BLB	0.85	0.93	0.89	88
Blast	0.92	0.82	0.87	94
Healthy	0.84	0.95	0.89	93
Hispa	0.98	0.96	0.97	85
Leaf Spot	0.96	0.84	0.89	80
Accuracy	0.90			
Macro Avg	0.91	0.90	0.90	440
Weighted Avg	0.90	0.90	0.90	440
VGG-19 - Fold 3				
BLB	0.86	0.92	0.89	88
Blast	0.89	0.87	0.88	94
Healthy	0.88	0.87	0.88	93
Hispa	0.91	0.91	0.91	85
Leaf Spot	0.87	0.84	0.85	80
Accuracy	0.88			
Macro Avg	0.88	0.88	0.88	440
Weighted Avg	0.88	0.88	0.88	440
VGG-19 - Fold 4				
BLB	0.85	0.95	0.90	88
Blast	0.90	0.84	0.87	94
Healthy	0.82	0.88	0.85	93
Hispa	0.94	0.92	0.93	85
Leaf Spot	0.86	0.75	0.80	80
Accuracy	0.87			
Macro Avg	0.87	0.87	0.87	440
Weighted Avg	0.87	0.87	0.87	440
VGG-19 - Fold 5				
BLB	0.94	0.77	0.85	88
Blast	0.86	0.88	0.87	94
Healthy	0.91	0.94	0.92	93
Hispa	0.78	0.94	0.85	85
Leaf Spot	0.96	0.88	0.92	80
Accuracy	0.88			
Macro Avg	0.89	0.88	0.88	440
Weighted Avg	0.89	0.88	0.88	440

For healthy samples, the model displayed minimal activation, demonstrating its confidence in identifying the absence of stress without being misled by natural thermal variations as shown in Fig. 6. This is a positive indication of the model's reliability.

Conversely, the Hispa group which was linked to insect infestation had acutely localized areas of activation as indicated in Fig. 7. This demonstrates the sensitivity to small-scale disruptions that are normally brought about by feeding damage or lesions by the

Table 4 Performance Metrics for ResNet-50 - Fold 1 to Fold 5

ResNet-50 - Fold 1				
Class	**Precision**	**Recall**	**F1-Score**	**Support**
BLB	0.95	0.94	0.95	88
Blast	0.96	0.97	0.96	93
Healthy	0.98	0.94	0.96	93
Hispa	0.99	0.97	0.98	86
Leaf Spot	0.87	0.94	0.90	80
Accuracy	0.95			
Macro Avg	0.95	0.95	0.95	440
Weighted Avg	0.95	0.95	0.95	440
ResNet-50 - Fold 2				
BLB	0.95	0.94	0.95	88
Blast	0.96	0.97	0.96	93
Healthy	0.98	0.94	0.96	93
Hispa	0.99	0.97	0.98	86
Leaf Spot	0.87	0.94	0.90	80
Accuracy	0.95			
Macro Avg	0.95	0.95	0.95	440
Weighted Avg	0.95	0.95	0.95	440
ResNet-50 - Fold 3				
BLB	0.92	1.00	0.96	88
Blast	0.93	0.91	0.92	94
Healthy	0.92	0.94	0.93	93
Hispa	0.96	0.96	0.96	85
Leaf Spot	0.96	0.86	0.91	80
Accuracy	0.94			
Macro Avg	0.94	0.94	0.94	440
Weighted Avg	0.94	0.94	0.94	440
ResNet-50 - Fold 4				
BLB	0.98	0.95	0.97	88
Blast	0.96	0.86	0.91	94
Healthy	0.84	0.96	0.89	93
Hispa	0.92	0.99	0.95	85
Leaf Spot	0.93	0.85	0.89	80
Accuracy	0.92			
Macro Avg	0.93	0.92	0.92	440
Weighted Avg	0.93	0.92	0.92	440
ResNet-50 - Fold 5				
BLB	0.96	0.88	0.92	88
Blast	0.86	0.95	0.90	94
Healthy	0.96	0.96	0.96	93
Hispa	0.95	0.95	0.95	85
Leaf Spot	0.96	0.94	0.95	80
Accuracy	0.93			
Macro Avg	0.94	0.93	0.93	440
Weighted Avg	0.94	0.93	0.93	440

model. Another disease BLB with its unpredictable impact on the physiology of leaves caused a wider and more asymmetric reaction, particularly around the veins and margins, which is expected in biology (as in Fig. 8). The heatmaps of the Blast disease had diffuse more centrally centered areas, which is indicative of extensive tissue integrity damage as demonstrated in Fig. 9.

Table 5 Performance Metrics for Xception - Fold 1 to Fold 5

Xception - Fold 1				
Class	**Precision**	**Recall**	**F1-Score**	**Support**
BLB	0.89	0.94	0.92	88
Blast	0.90	0.88	0.89	93
Healthy	0.89	0.92	0.91	93
Hispa	0.97	0.98	0.97	86
Leaf Spot	0.89	0.80	0.84	80
Accuracy	0.91			
Macro Avg	0.91	0.91	0.91	440
Weighted Avg	0.91	0.91	0.91	440
Xception - Fold 2				
BLB	0.95	0.95	0.95	88
Blast	0.96	0.81	0.88	94
Healthy	0.82	0.96	0.88	93
Hispa	0.97	0.99	0.98	85
Leaf Spot	0.94	0.90	0.92	80
Accuracy	0.92			
Macro Avg	0.93	0.92	0.92	440
Weighted Avg	0.93	0.92	0.92	440
Xception - Fold 3				
BLB	0.95	0.94	0.95	88
Blast	0.92	0.94	0.93	94
Healthy	0.91	0.89	0.90	93
Hispa	0.93	0.95	0.94	85
Leaf Spot	0.87	0.86	0.87	80
Accuracy	0.92			
Macro Avg	0.92	0.92	0.92	440
Weighted Avg	0.92	0.92	0.92	440
Xception - Fold 4				
BLB	0.94	0.94	0.94	88
Blast	0.90	0.91	0.91	94
Healthy	0.95	0.90	0.93	93
Hispa	0.92	0.96	0.94	85
Leaf Spot	0.90	0.89	0.89	80
Accuracy	0.92			
Macro Avg	0.92	0.92	0.92	440
Weighted Avg	0.92	0.92	0.92	440
Xception - Fold 5				
BLB	0.98	0.97	0.97	88
Blast	0.95	0.99	0.97	94
Healthy	0.94	0.91	0.93	93
Hispa	0.98	0.99	0.98	85
Leaf Spot	0.92	0.91	0.92	80
Accuracy	0.95			
Macro Avg	0.95	0.95	0.95	440
Weighted Avg	0.95	0.95	0.95	440

By comparison, the leaf spot heatmap overlay reveals high intensity of the vertical structures of the leaf. This means that the model would be selective in terms of which textures or patterns it would see on the leaf blade to recognize the disease instead of the black background or the unnecessary noise as illustrated in Fig. 10.

These visual differences by the class not only confirmed the ability of the model to differentiate between different types of stress but also demonstrated how thermal imaging

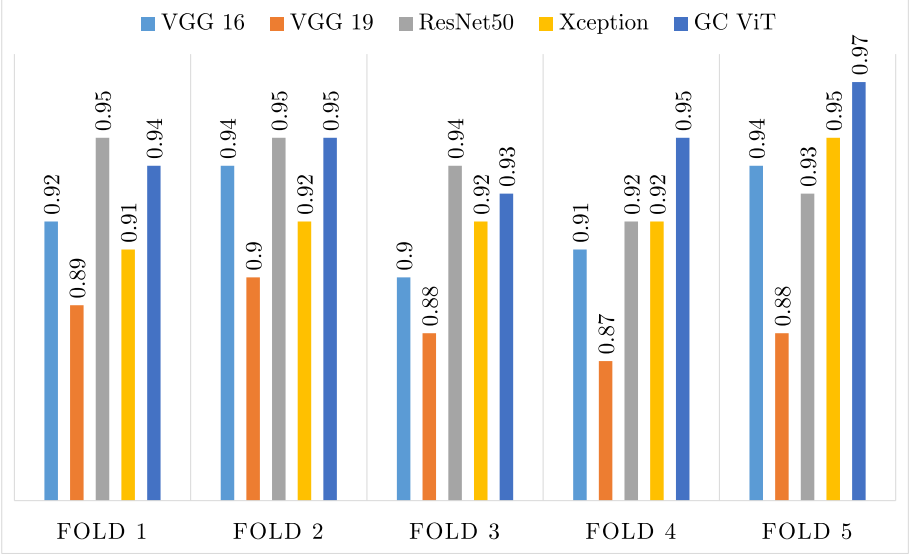


Fig. 5 Accuracy of models over fold

with the help of interpretable AI can reveal the physiological signature of each type of stress. Finally, XAI has also turned our model into a transparent diagnostic instrument, which is a sign that in the complex language of plant stress, the GCViT model is able not only to detect but also to understand. It can be concluded that ResNet-50 and GCViT are the best models in terms of accuracy and consistency on all folds and classes. Their good performance qualifies them to be the best candidates when it comes to applications that need good performance and reliable classification systems. Also, further investigation of the underperformance of the class of leaf spot could have provided the enhancements in all models and increase their overall performance. Transformer computational and memory costs are higher than CNNs, and this creates latency and energy constraints to resource-constrained IoT devices, which are prevalent in the agricultural field. This added complexity is, however, worth it to be able to detect the plant stresses as Transformers can capture the global context in order to detect the non-localized but subtle physiological changes that cannot be effectively identified using the conventional CNN filters.

5 Conclusion

Findings in this work show that it is important to choose appropriate deep learning models based on the task and the type of data. This study suggests that GCViT is an effective model for detecting stress in paddy plants using thermal imaging. GCViT achieved high accuracy rates between 0.93 and 0.97, with macro and weighted average F1-scores consistently above 0.94. Its performance varied by only about 3% across different dataset folds. Integrating explainable artificial intelligence (XAI) with the GCViT model, helps interpretability of the model and decision making in identifying plant stresses using thermal images. The class-specific heatmaps showed that the model focused on the thermal areas related to different stress levels, while showing low activation in healthy samples. This clarity not only shows that the model is reliable but also helps apply it in precision agriculture for accurate stress diagnosis. In the future, model's speed and complexity in terms of trainable parameters can be improved for real-time applications.

Table 6 Performance Metrics for GCViT - Fold 1 to Fold 5

GCViT - Fold 1				
Class	**Precision**	**Recall**	**F1-Score**	**Support**
BLB	0.97	0.99	0.98	88
Blast	0.90	1.00	0.95	93
Healthy	0.93	0.86	0.89	93
Hispa	0.98	0.95	0.96	86
Leaf Spot	0.94	0.90	0.92	80
Accuracy	0.94			
Macro Avg	0.94	0.94	0.94	440
Weighted Avg	0.94	0.94	0.94	440
GCViT - Fold 2				
BLB	0.97	0.95	0.96	88
Blast	0.94	0.96	0.95	94
Healthy	0.92	0.96	0.94	93
Hispa	0.98	0.96	0.97	85
Leaf Spot	0.99	0.94	0.96	80
Accuracy	0.95			
Macro Avg	0.96	0.95	0.96	440
Weighted Avg	0.96	0.95	0.95	440
GCViT - Fold 3				
BLB	0.97	0.99	0.98	88
Blast	0.92	0.97	0.94	94
Healthy	0.88	0.90	0.89	93
Hispa	0.95	0.93	0.94	85
Leaf Spot	0.93	0.85	0.89	80
Accuracy	0.93			
Macro Avg	0.93	0.93	0.93	440
Weighted Avg	0.93	0.93	0.93	440
GCViT - Fold 4				
BLB	0.97	0.98	0.97	88
Blast	0.95	0.96	0.95	94
Healthy	0.91	0.95	0.93	93
Hispa	0.98	0.98	0.98	85
Leaf Spot	0.93	0.86	0.90	80
Accuracy	0.95			
Macro Avg	0.95	0.94	0.94	440
Weighted Avg	0.95	0.95	0.95	440
GCViT - Fold 5				
BLB	0.99	0.94	0.97	88
Blast	0.96	0.97	0.96	94
Healthy	0.97	0.96	0.96	93
Hispa	0.94	1.00	0.97	85
Leaf Spot	0.97	0.96	0.97	80
Accuracy	0.97			
Macro Avg	0.97	0.97	0.97	440
Weighted Avg	0.97	0.97	0.97	440

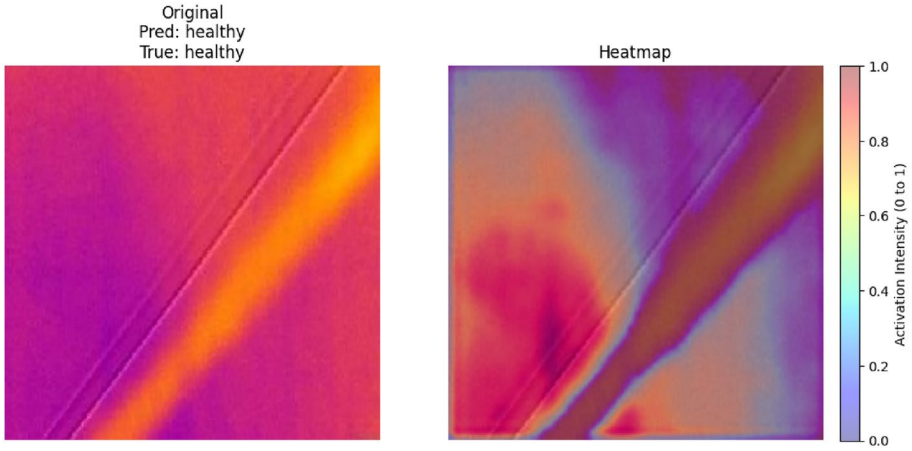


Fig. 6 Original image and Heatmap for class healthy

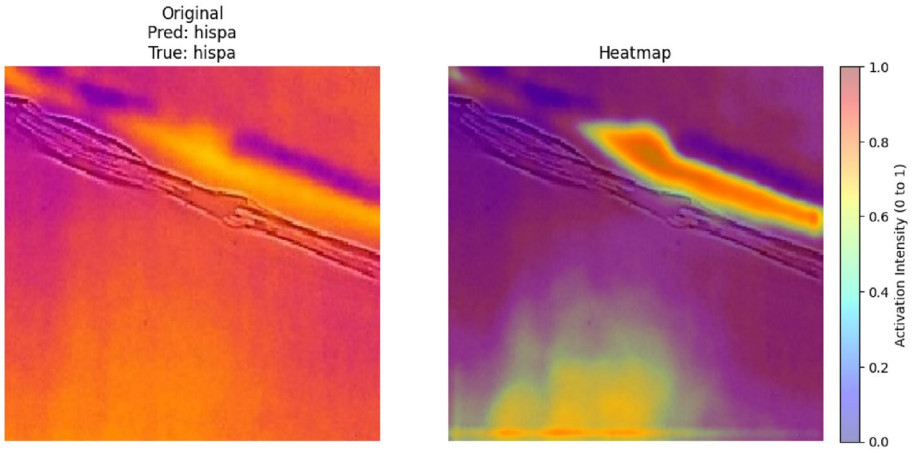


Fig. 7 Original image and Heatmap for class hispa

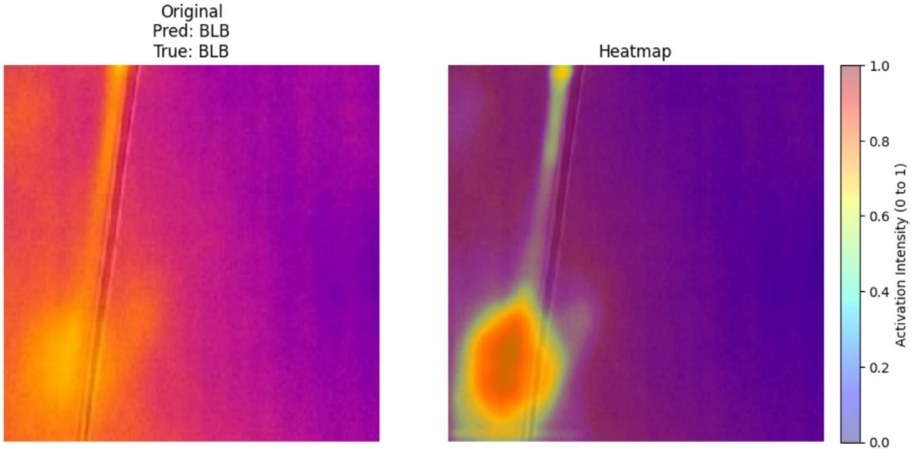


Fig. 8 Original image and Heatmap for class BLB

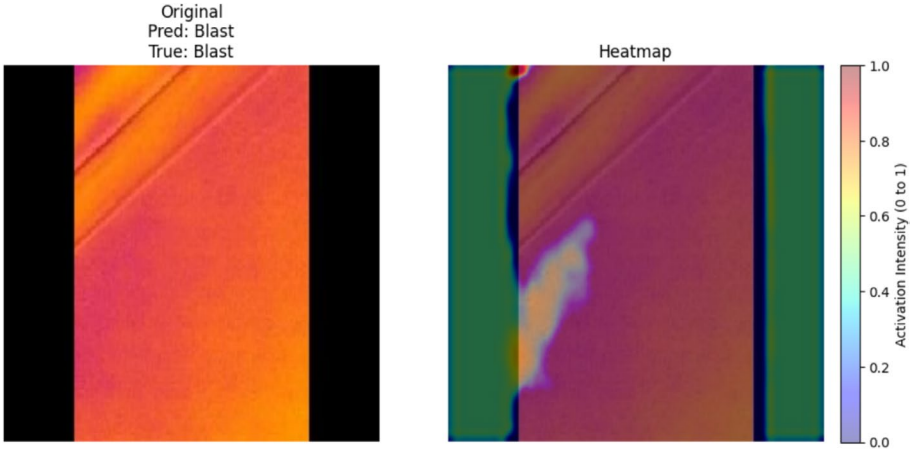


Fig. 9 Original image and Heatmap for class Blast

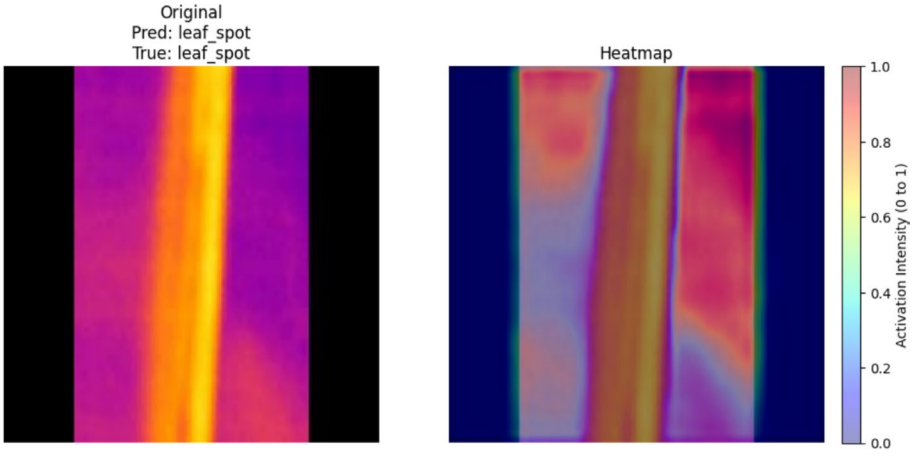


Fig. 10 Original image and Heatmap for class leaf spot

Abbreviations

AI	Artificial intelligence
ANN	Artificial neural network
CNN	Convolutional neural network
DL	Deep learning
GC-ViT	Global context vision transformer
Grad-CAM	Gradient-weighted class activation mapping
LSTM	Long short-term memory
MLP	Multilayer perceptron
NLP	Natural language processing
ReLU	Rectified linear unit
ResNet50	Residual network (50 layers)
ROC	Receiver operating characteristic
SVM	Support vector machine
VGG16	Visual geometry group (16 layers)
VGG19	Visual geometry group (19 layers)
XAI	Explainable artificial intelligence

Author contributions

S.P.: Conceptualization, methodology implementation, result analysis, and original draft preparation. S.K.: Conceptualization, supervision, and original draft preparation.

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Data availability
The datasets analyzed during the current study are publicly available in the Kaggle repository and are cited in the Dataset section.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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